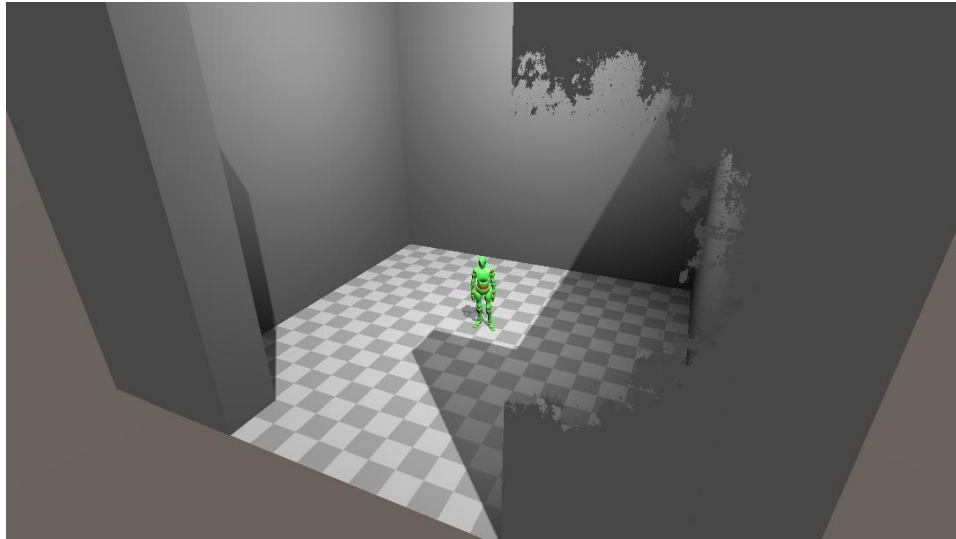


PixelPulse – Documentation



Dynamic Occlusion Cutout System (DOCS) - SAMPLE

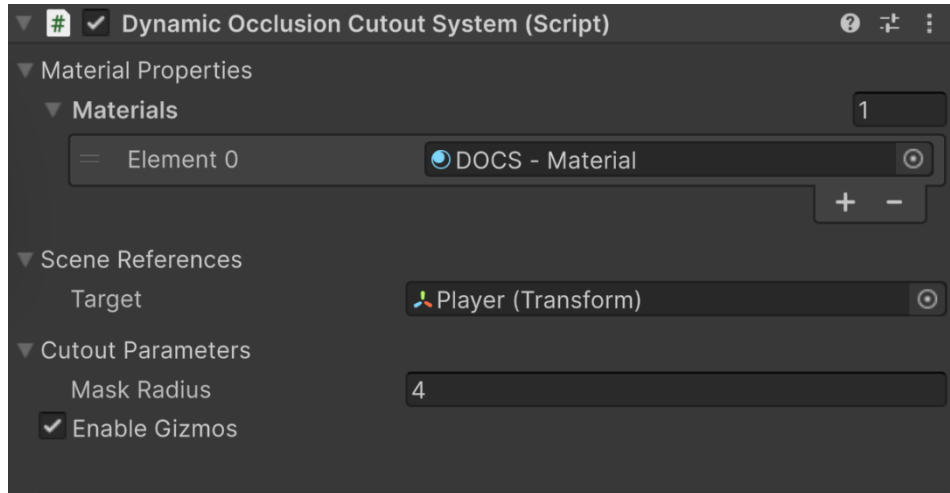
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Script

This script is used to **calculate** and **update** the Dynamic Occlusion Cutout System (DOCS) **Materials**.

It has the following parameters:



Material Properties

- **Materials** are the references to the DOCS materials to update.

Scene References

- **Target** is the aim of the camera; it should stand behind the objects that will be cut out.

Cutout Parameters

⚠ **The DOCS – SAMPLE offers only Physical Detection** ⚠

It will deliver the most precise results, but it requires the use of objects that have physical components on them (Collider or Trigger).

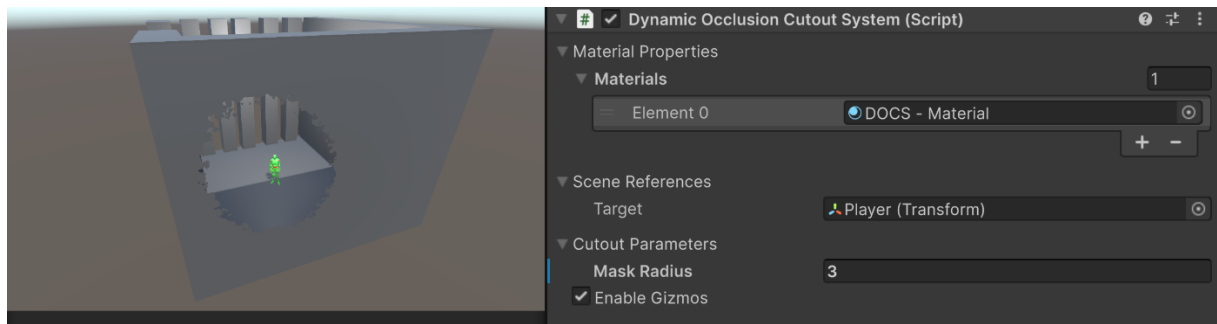
Comparison Table

Feature	Physical	Graphical	Constant
Uses colliders	✓	✗	✗
Uses renderers	✗	✓	✗
Occlusion-aware	✓	✓	✗
Physics-based	✓	✗	✗
Adaptive radius support	✓ ¹	✓	✓
Performance cost	Low	Medium-High	Very Low
Visual-only scenes	✗	✓	✓
Deterministic output	⚠ ²	✓	✓
Unity Terrain (foliage & trees)	✗	✓	✗

¹ Adaptive Radius support is only available in the DOCS Pro

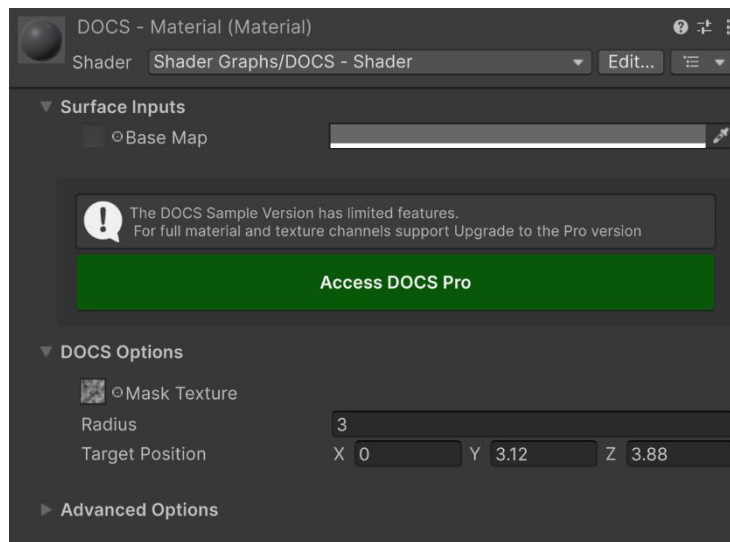
² Dependent of Unity's Physics parameters configuration

- **Mask Radius** is the radius of the cutout mask send to the DOCS materials.



Shader

The shader enables the creation of DOCS Materials.



The sample material supports the **Base Map & Base Color**, and the DOCS parameters:

- **Target Position** is updated by the script (can be changed manually for debugging in editor)
- **Radius** is updated by the script (can be changed manually for debugging in editor)
- **Mask Texture** is the texture used for the occlusion cutout.

! ! !

MaskTextures use Triplanar UVs, this setting can be changed in the shader but first be sure YOU KNOW WHAT YOU ARE DOING if you want to change its functionality

! ! !

How to use it?

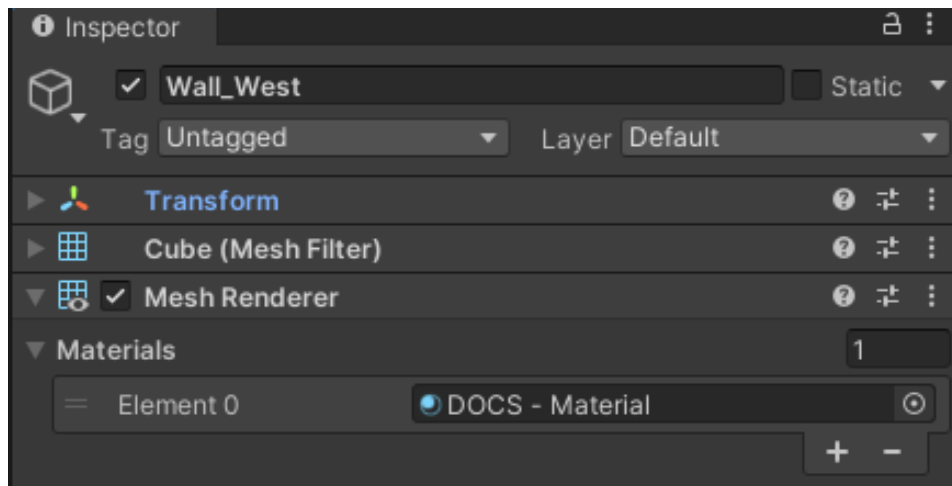
To use this system:

- **Create a material** derived from the **DOCS – Shader**

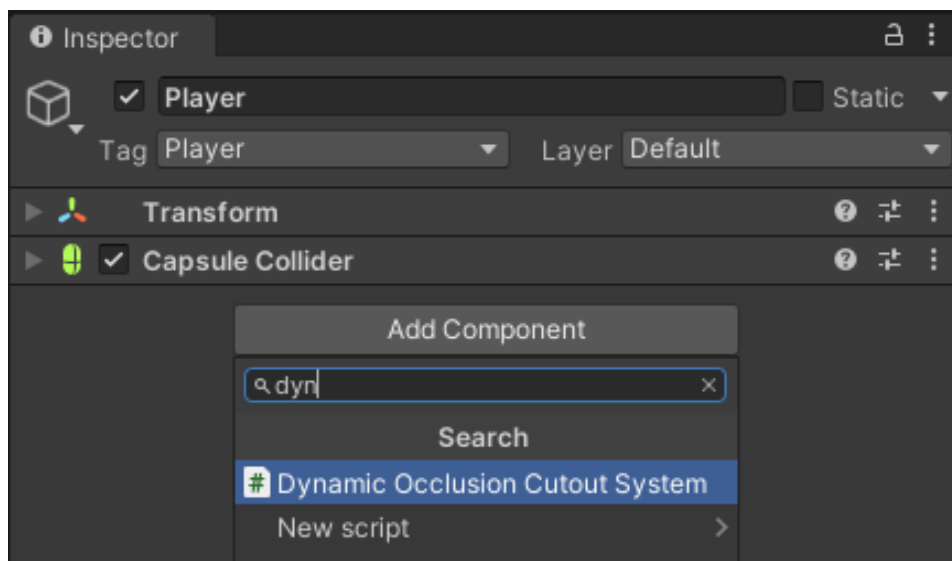


- **Right click** on the shader > **Create** > **Material**
In Assets > Dynamic Occlusion Cutout System (DOCS) > Shader

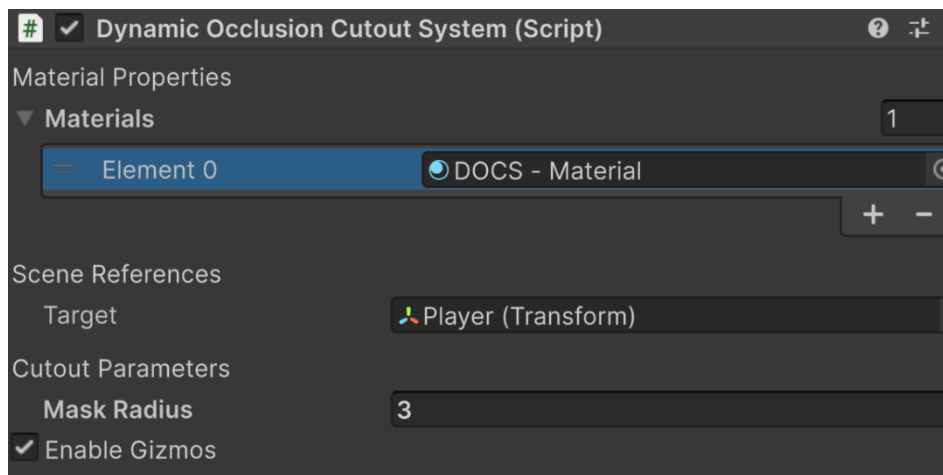
- **Place the Material** on the GameObjects you want to see through (ex. Walls)
Inspector > Mesh Renderer > Materials > Element (n)



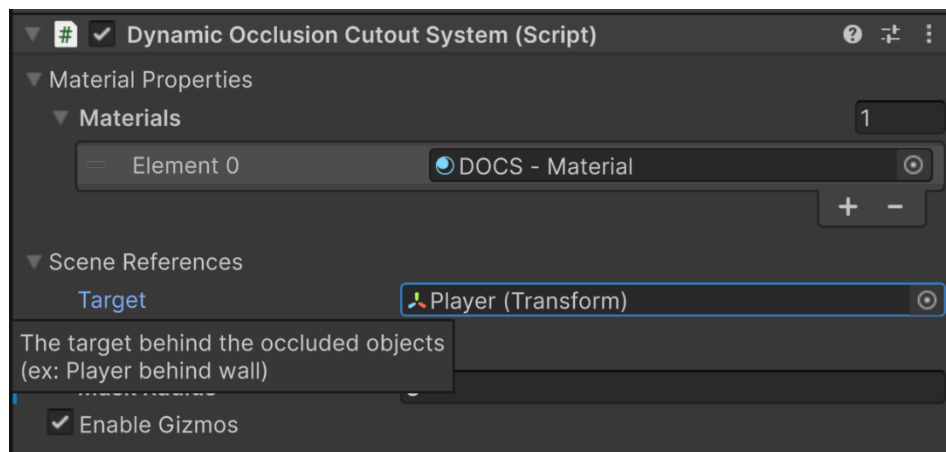
- **Assign the Dynamic Occlusion Cutout System - Script** on any object in your scene
The most relevant object to hold the system is generally the character or camera (any object can hold it; it should still work).



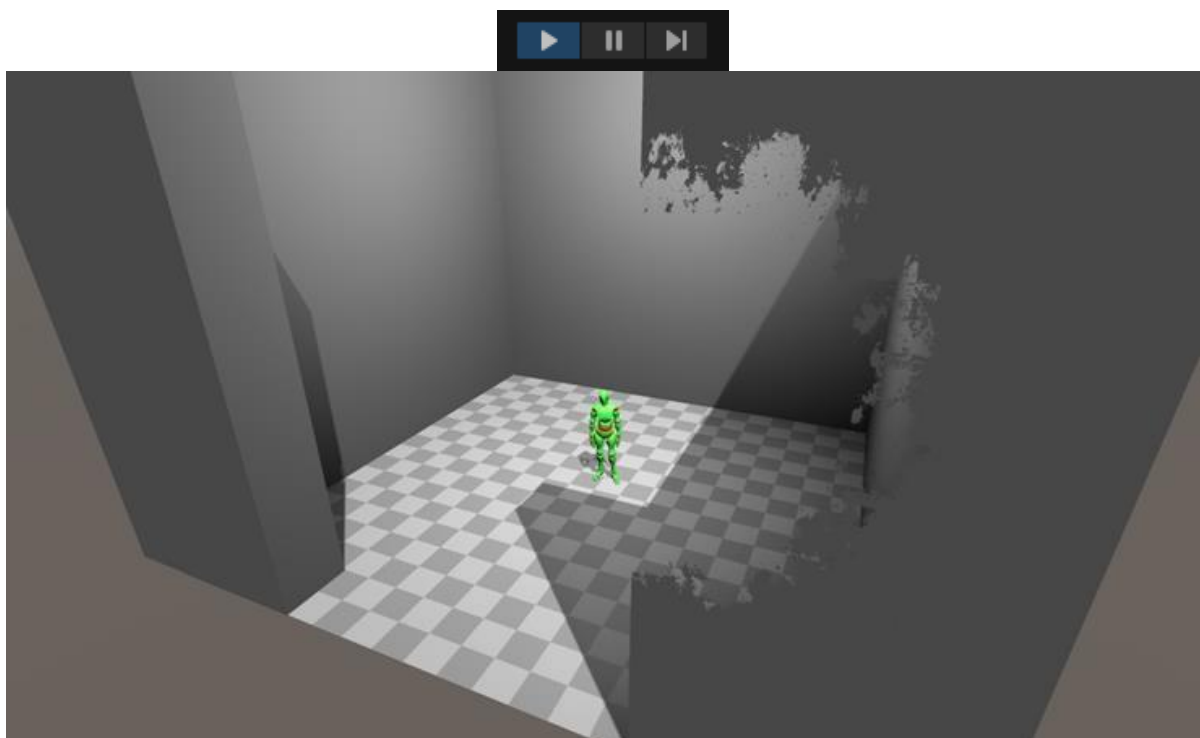
- **Assign** the newly created **material** in the script for it to be updated



- **Assign the Target**

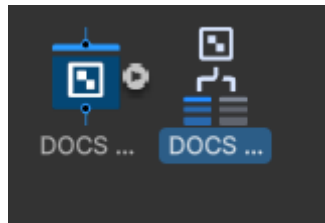


- Finally click **Play** and **Enjoy** 😊

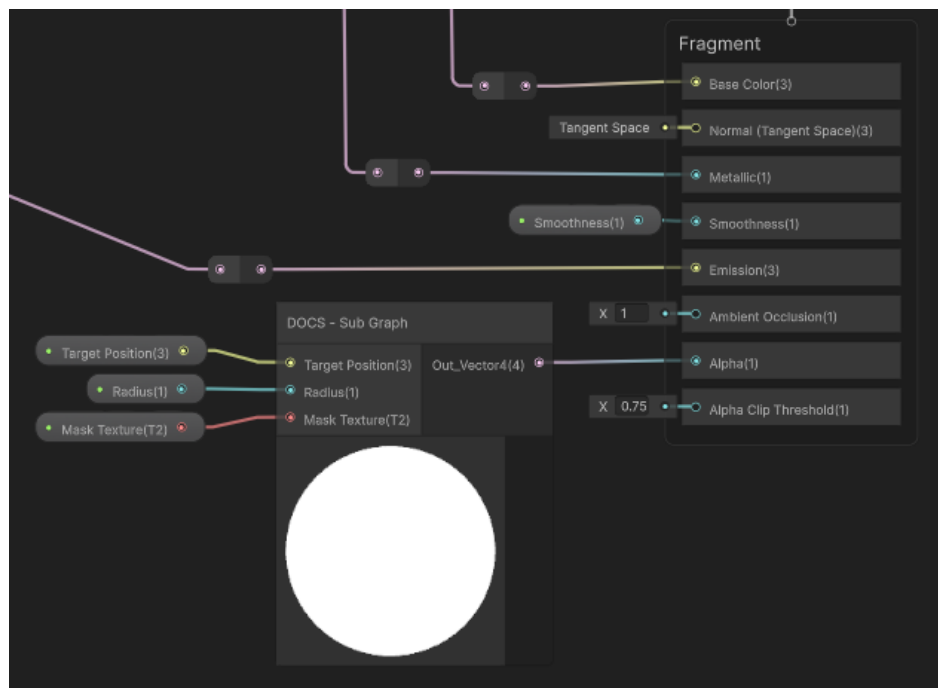


Integration with existing materials

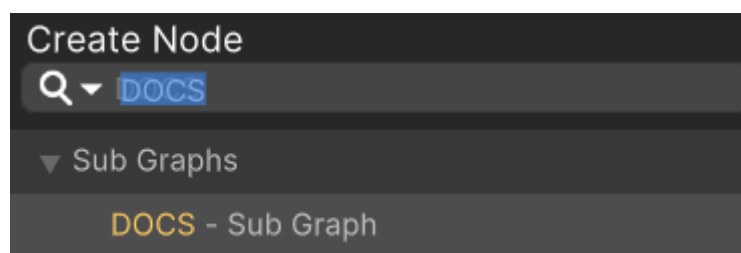
To facilitate the integration with already existing materials the version 1.0.2 introduced a Sub-Graph.



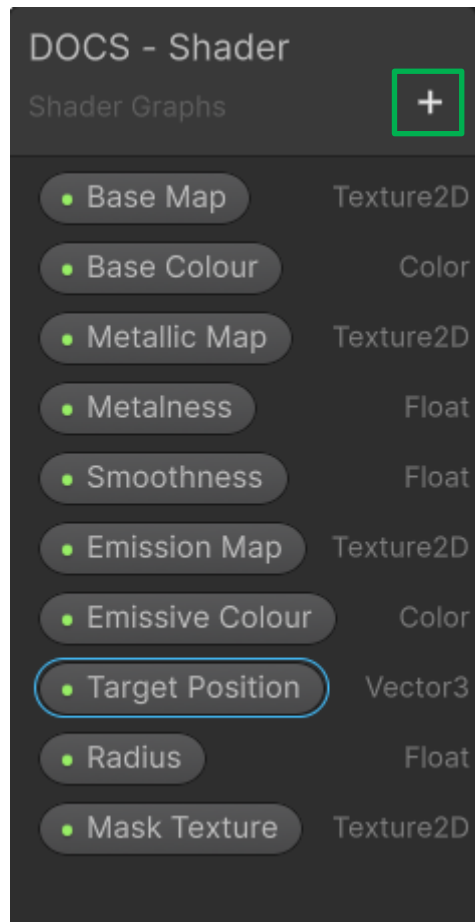
To use it, it must be integrated into the shader used for walls/elements you want to cut out



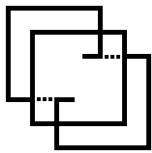
To do so, create a new node in the ShaderGraph (Right click) and search for “**DOCS**” and link it to the alpha channel.



Then, declare a **Vector3** called “**Target Position**”, a **float** called “**Radius**” and link them to the node.



Optionally, you can declare and change the “**Mask Texture**” for a different style of cut out. Once set up, the modified materials should behave like the DOCS materials.



The PixelPulse team thanks you for purchasing
Dynamic Occlusion Cutout System - DOCS
and wishes you all the best for your project.
